Test plan and evidence

SignFlow — Digital Document Approval Workflow for a University

1. Test objectives

The test effort checks that files are routed correctly, approval actions are recorded, signatures are handled through the selected provider, reminders work, and no submitted file becomes unretrievable. The decision table focuses on file type and amount because those values drive routing.

2. Test levels and entry/exit criteria

Levels: unit/configuration tests, integration tests, system tests, security and audit checks, performance checks, and user acceptance testing.

Entry criteria: approved requirements baseline, configured top-three rules, test users, test signature credentials or sandbox, and test data.

Exit criteria: all Must tests pass; no open critical defect; legal/signature and audit checks pass; lost-file check passes; failed tests have an owner and retest result.

3. Boundary-value analysis

The brief asks for BVA of approval-level count and reminder interval. Exact allowed ranges are not supplied, so the following are explicit validation assumptions to be confirmed during requirements review.

3.1 Approval-level count

Assumption for test design: allowed range is 1 to 10 inclusive.

Test ID	Input	Expected result
BVA-AL-01	0	Reject; below minimum
BVA-AL-02	1	Accept; minimum valid value
BVA-AL-03	2	Accept; just above minimum
BVA-AL-04	9	Accept; just below maximum
BVA-AL-05	10	Accept; maximum valid value
BVA-AL-06	11	Reject; above maximum

3.2 Reminder interval

Assumption for test design: allowed range is 1 to 30 days inclusive.

Test ID	Input	Expected result
BVA-RI-01	0 days	Reject; below minimum
BVA-RI-02	1 day	Accept; minimum valid value
BVA-RI-03	2 days	Accept; just above minimum
BVA-RI-04	29 days	Accept; just below maximum
BVA-RI-05	30 days	Accept; maximum valid value
BVA-RI-06	31 days	Reject; above maximum

If the Registrar approves different limits, these cases should be updated before execution rather than silently changing the expected result.

4. Equivalence classes

Field	Valid classes	Invalid classes
File type	Purchase request, leave, event approval, configured other type	Blank, unsupported type
Amount	Zero or positive amount where applicable; amount in a configured band	Negative, non-numeric, missing where mandatory
Approval-level count	Integer 1–10 (assumption)	0, 11+, decimal, blank, text
Reminder interval	Integer 1–30 days (assumption)	0, 31+, decimal, blank, text
Signature result	Validated, pending	Invalid, expired, provider error

5. Decision table for routing

The case does not provide exact amount thresholds or role names. The table below therefore shows the structure required for the rules; the threshold labels are configuration placeholders and must be replaced by the approved university values before release.

Rule	File type	Amount condition	Expected route
R1	Purchase request	Low band	Section officer → authorised approver
R2	Purchase request	High band	Section officer → senior officer → authorised approver
R3	Leave	Any configured amount / no amount	Section officer → senior officer
R4	Event approval	Low band	Section officer → event approver
R5	Event approval	High band	Section officer → senior officer → event approver
R6	Unsupported/other	Any	Exception queue; no automatic approval route

Decision-table tests: TC-10 through TC-15 cover R1–R6. TC-16 checks a boundary just below an approved threshold, and TC-17 checks the threshold and just above it after the Registrar supplies the actual values.

6. Test cases

ID	Test	Expected result
TC-01	Create valid purchase request with attachment	Unique file ID and Draft status are created.
TC-02	Submit valid leave file	File moves to Submitted and is retrievable by ID.
TC-03	Submit without file type	Submission is rejected with a field-level error.
TC-04	Submit with unsupported attachment metadata	Submission is rejected and no route is created.
TC-05	Open assigned inbox item	Current owner, level, status, and due date are shown.
TC-06	Approve a review step	Decision, actor, timestamp, and next state are recorded.
TC-07	Return a file with reason	File moves to Returned and requester sees the reason.
TC-08	Request signature	Signature provider reference and Pending status are stored.
TC-09	Receive valid signature result	Signature becomes Validated and approval continues.
TC-10	Purchase request in low amount band	Route matches R1.
TC-11	Purchase request in high amount band	Route matches R2.
TC-12	Leave file	Route matches R3.
TC-13	Event approval in low amount band	Route matches R4.

ID	Test	Expected result
TC-14	Event approval in high amount band	Route matches R5.
TC-15	Unsupported file type	File enters exception queue under R6.
TC-16	Amount immediately below approved threshold	Lower-band route is selected.
TC-17	Amount at and immediately above approved threshold	Approved threshold route is selected and no ambiguity exists.
TC-18	Reminder interval reached	Reminder event is created and notification is sent.
TC-19	Due date exceeded	Escalation event is created for the configured owner.
TC-20	Complete a file lifecycle	Audit contains all required events and status changes.
TC-21	Search by ID, type, status, owner	Results contain only matching files.
TC-22	Generate approval-time report	Report includes period, average, open, overdue, and exception values.
TC-23	Validate signature method approval record	Written legal/compliance approval is present before release.
TC-24	Inspect audit records for all test transactions	100% contain required fields.
TC-25	Run availability measurement in agreed window	Monthly result meets 99.5% target, excluding approved maintenance.
TC-26	Run response-time sample	At least 95% of normal page requests complete within 3 seconds.
TC-27	Calculate approval-time result	Average is under 4 days for the agreed reporting period.

ID	Test	Expected result
TC-28	Retrieve every submitted test file	Zero submitted files are unretrievable.
TC-29	Attempt unauthorised access	Access is denied and the attempt is logged.

7. Defect log and example evidence

The case does not provide an executed test log. The table below is a compact evidence template with example entries to show how DRE and defect density are calculated. Replace the example counts with actual execution results before claiming project metrics.

Defect ID	Found in	Severity	Description	Status
D-01	Integration test	High	High-band purchase request selected the low-band route.	Closed and retested
D-02	System test	Medium	Reminder was sent one day late at the configured boundary.	Closed and retested
D-03	UAT	Medium	Returned file did not show the return reason in requester view.	Closed and retested
D-04	UAT	High	Signature validation failure left the file looking approved.	Closed and retested
D-05	UAT	Low	Search by current owner used case-sensitive matching.	Closed and retested

8. Metrics

8.1 Defect removal efficiency

Using the five example defects above:

- Defects found before release = 5.
- Defects found after release = 0 in this illustrative dataset.

[DRE = \frac{\text{defects found before release}}{\text{defects found before release} + \text{defects found after release}} \times 100]

[DRE = \frac{5}{5 + 0} \times 100 = 100\%]

This 100% is an example based on the table, not a measured production result.

8.2 Defect density

Assumption for this evidence template: 12 file-type configurations are the size unit because the case supplies 12 file types and no lines-of-code or function-point count.

[\text{Defect density} = \frac{5\text{ defects}}{12\text{ file-type configurations}} = 0.4167\text{ defects/configuration}]

Report the unit explicitly. Do not compare this number with a LOC-based defect-density benchmark without converting the denominator.

9. Evidence to retain

- Test execution record with tester, environment, date, input, expected result, actual result, and status.
- Screenshots or exports for route selection and audit history.
- Signature provider validation response and legal/compliance approval.
- Reminder and escalation notification records.
- Approval-time report and lost-file reconciliation.
- Defect retest results and release decision.